
FedReCall: Recalling Client-Specific Directions in Federated LoRA Fine-tuning

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Abstract

1 Low-rank adaptation (LoRA) enables communication-efficient federated fine-
2 tuning of large language models, but standard FedLoRA averages the factors
3 A and B separately even though the model update is determined by their product
4 $\Delta W = sBA$. Under task heterogeneity, factor-wise aggregation can further dilute
5 directions that are locally useful but not strongly shared across clients; we call
6 this phenomenon *directional dilution*. We analyze this effect through the singular
7 directions of client LoRA updates and find that dilution is concentrated on low-
8 strength directions: local training repeatedly restores them, aggregation suppresses
9 them again, and their recovery is stronger for the same client than for other clients,
10 suggesting client-specific adaptation signals rather than random noise. We propose
11 FEDRECALL, a client-private plug-in that estimates observable dilution gaps after
12 aggregation, filters reliable low-strength directions with a quality gate, stores them
13 in a private frozen cache, and recalls them through a low-rank bypass when the
14 client next participates. As a plug-in, FEDRECALL can be seamlessly integrated
15 into existing FedLoRA methods without changing their aggregation rules, com-
16 munication payloads, or trainable-parameter sets, improving both personalized
17 performance and global evaluation metrics with low additional overhead.

18 1 Introduction

19 Large language models (LLMs) are increasingly adapted to specialized tasks, but full fine-tuning
20 remains expensive at scale [32, 9]. Low-rank adaptation (LoRA) [12] offers a practical alternative by
21 training a compact update $\Delta W = \frac{\alpha}{r}BA$ while keeping the backbone frozen. When task data is dis-
22 tributed across privacy-sensitive clients, federated LoRA (FedLoRA) further reduces communication
23 because clients exchange only lightweight adapter factors rather than full model weights [23, 8].

24 Existing FedLoRA methods mainly improve how LoRA factors are aggregated or decide which
25 components remain private [39, 30, 3, 34, 2, 15, 28]. Personalized FL offers another route, but
26 typical PFL designs introduce extra local modules, optimization stages, or personalization-specific
27 structures [1, 31, 17, 27]. These approaches improve different parts of the overhead–capability
28 trade-off, as summarized in Table 1, but they do not directly ask which directions aggregation actually
29 erases.

30 Federated learning compresses multiple client objectives into a single shared update, so task
31 heterogeneity naturally creates *update conflict*: directions with broad cross-client support tend
32 to dominate, while directions that are useful to one client but weakly shared are more likely
33 to be suppressed. In FedLoRA, this conflict is amplified by factor-wise aggregation. The
34 server averages A and B separately, but model behavior is governed by the effective update
35 $\Delta W = sBA$; therefore, aggregation can retain different local directions to very different de-
36 grees. We call the resulting selective attenuation of locally useful directions *directional dilution*.

Our key insight is that the directions most vulnerable to aggregation are not arbitrary noise, but recurring client-specific adaptation signals. As shown in Figure 2, these directions suffer larger relative dilution, are repeatedly recovered by local training and suppressed again by aggregation, and correspond to the fragile client-specific components that standard sharing tends to smooth, distort, or cancel (Figure 2c). Existing methods do not explicitly identify and locally restore this diluted directional signal.

We propose *FedReCall* (*Federated Re-injection of Client-Adaptive Lost directions*) to answer this question with a client-private recall mechanism. After each aggregation, a participating client decomposes its effective LoRA update, measures how much the new global update retains along each local direction, and converts the missing positive strength into an observable dilution gap. Reliable diluted directions are stored as a private frozen cache and recalled as a low-rank bypass when the client next participates; a single-batch quality gate enables the bypass only when it improves the client’s local loss. The cache is never uploaded, so clients still communicate only clean LoRA factors and the server runs the original aggregation rule unchanged. FedReCall therefore adds personalized memory without extra trainable parameters, communication payload, or a new server-side optimizer.

Mechanism diagnostics and main experiments confirm that this local recall reduces directional dilution, dampens the recovery–disruption cycle, and improves both personalized client quality and global evaluation metrics across representative FedLoRA aggregators (Figure 1). As Table 1 highlights, it does so as a plug-in without changing the aggregation rule, communication payload, or trainable-parameter set. Together, these results show that diluted client-specific directions can be recovered with low additional overhead, yielding stronger personalized and global performance while preserving the original federated training protocol.

Contributions.

- We uncover directional dilution, a structured form of personalization loss where factor-wise aggregation repeatedly weakens client-specific directions in the effective LoRA update space.
- We propose FedReCall, a client-private recall plug-in that restores diluted directions with an observable dilution gap, a frozen low-rank bypass, and a lightweight quality gate.
- We show that FedReCall improves personalized and global performance across representative FedLoRA aggregators while preserving the original server rule, communication payload, and trainable-parameter set.

2 Related Work

Parameter-Efficient Fine-Tuning Parameter-efficient fine-tuning (PEFT) aims to adapt a large pre-trained model to downstream tasks while updating only a small subset of task-specific parameters and keeping the backbone largely frozen[9]. A variety of PEFT techniques have been explored, including prompt-based tuning[18, 16, 14], ladapter-based methods[11], low-rank adaptation methods such as LoRA[12, 22, 24], and so on. In this paper, we focus on LoRA, as it has become one of the most widely used PEFT methods for large models due to its strong adaptation performance, parameter efficiency, and broad applicability across downstream tasks[12, 8, 7, 21]. While we adopt LoRA’s structural assumptions, our focus is not on LoRA in isolation but on a federated-specific failure

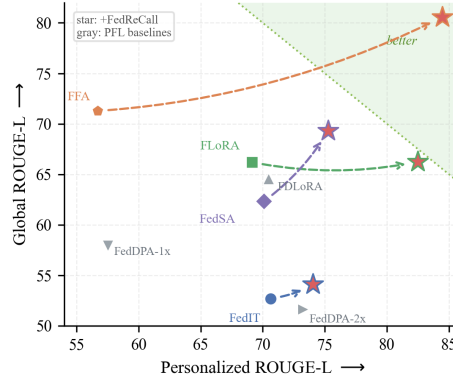


Figure 1: Performance balance between client-side and global performance.

Table 1: Prerequisites and application scope of FedLoRA methods.

Method	Extra-param free	Extra-optm free	Extra-comm free	Personalization	Plug-in
FedIT	✓	✓	✓	✗	–
FFA-LoRA	✓	✓	✓	✗	✗
FedSA-LoRA	✓	✓	✓	✗	✗
FLoRA	✓	✓	✗	✗	✗
FDLoRA	✗	✗	✓	✓	✗
FedDPA-T	✗	✗	✓	✓	✗
FedReCall	✓	✓	✓	✓	✓

mode that emerges only when the bilinear LoRA product BA is split across clients and aggregated factor-by-factor.

Parameter-Efficient Fine-Tuning in Federated Learning Recent work has introduced PEFT into federated learning to make the fine-tuning of pre-trained or foundation models feasible under limited client-side computation, memory, and communication budgets, since only a small subset of parameters needs to be optimized and exchanged[29, 3]. Existing studies have explored several PEFT mechanisms in federated settings, including prompt-based tuning[20], adapter-style modules[37, 35], and LoRA-based fine-tuning[3, 40, 34, 2]. Among them, LoRA has become a particularly active direction in federated fine-tuning, with prior work addressing model heterogeneity and personalization[40], aggregation accuracy for heterogeneous LoRA adapters[4], privacy-preserving and more stable LoRA optimization[30, 13], and robustness under increasing data heterogeneity[15, 3].

Personalized Federated Learning Personalized federated learning (PFL) studies how to tailor federated models to individual clients under statistical heterogeneity. Because client data distributions often differ substantially, a single global model may not perform equally well for every client[1, 6, 31, 17]. Existing PFL methods address this issue in several ways. Some split the model into globally shared and client-specific components, such as keeping a shared backbone with local personalization layers[1, 25] or learning a shared representation with client-specific heads[5, 36]. Others learn personalized models together with shared knowledge, for example by adaptively combining local and global models[6, 10], or by learning personalized weighted combinations of available models[42], decoupling personalized optimization from global model learning in a bi-level formulation[31], or improving fairness and robustness through personalization[17, 19].

3 Method

3.1 Preliminaries and notations

Consider a federated learning system with N clients, each holding a private dataset $\mathcal{D}_c$. All clients share a pre-trained backbone W_{base} and fine-tune it via LoRA [12]: each LoRA-adapted layer l adds a low-rank update $\Delta W_l = s_l B_l A_l$, where $B_l \in \mathbb{R}^{d_{\text{out}} \times r}$, $A_l \in \mathbb{R}^{r \times d_{\text{in}}}$, and $s_l = \alpha_l/r$ is the LoRA scaling factor. In each communication round t , a subset $S_t \subseteq [N]$ of clients is selected. Each selected client c downloads the current global LoRA factors, performs local training, and uploads the trained factors $(A_{c,l}^t, B_{c,l}^t)$. Following standard personalized federated learning practice, we evaluate per-client loss on each client’s own data after a local adaptation step, with the global LoRA factors serving as a federated initialization rather than the deployment target.

The server aggregates LoRA factors using the standard factor-wise FedIT protocol:

$$A_l^{t,\text{glob}} = \sum_{j \in S_t} p_j A_{j,l}^t, \quad B_l^{t,\text{glob}} = \sum_{j \in S_t} p_j B_{j,l}^t, \quad \sum_{j \in S_t} p_j = 1. \quad (1)$$

This produces the actual global LoRA update used in the next round,

$$\Delta W_l^{t,\text{glob}} = s_l B_l^{t,\text{glob}} A_l^{t,\text{glob}}. \quad (2)$$

Because the LoRA product is bilinear, factor-wise averaging does not preserve the effective update:

$$s_l B_l^{t,\text{glob}} A_l^{t,\text{glob}} \neq \sum_{j \in S_t} p_j s_l B_{j,l}^t A_{j,l}^t \quad (3)$$

in general, so individual client directions can be selectively lost in the aggregated update even when both factors are faithfully averaged.

Direction decomposition. Each client’s effective LoRA update admits a singular value decomposition

$$\Delta W_{c,l}^t = \sum_{i=0}^{r-1} \sigma_i u_i v_i^\top, \quad (4)$$

where $\sigma_0 \geq \dots \geq \sigma_{r-1} \geq 0$ are the singular values (amplitudes) and $u_i \in \mathbb{R}^{d_{\text{out}}}$, $v_i \in \mathbb{R}^{d_{\text{in}}}$ are the corresponding singular vectors. We call each rank-one atom $\phi_i = u_i v_i^\top$ a *direction* in the update space; because u_i and v_i are columns of orthonormal bases, $\|\phi_i\|_F = 1$ and distinct atoms are Frobenius-orthogonal. The Frobenius inner product $\langle M, \phi_i \rangle_F = u_i^\top M v_i$ measures how much of direction ϕ_i is present in any matrix M . We group directions by index position $q_i = i/r$ into four bands: *head* ($q < 0.2$), *middle* ($0.2 \leq q < 0.6$), *tail* ($0.6 \leq q < 0.9$), and *floor* ($q \geq 0.9$). The head band carries the largest-amplitude directions; the floor band carries the weakest non-trivial ones.

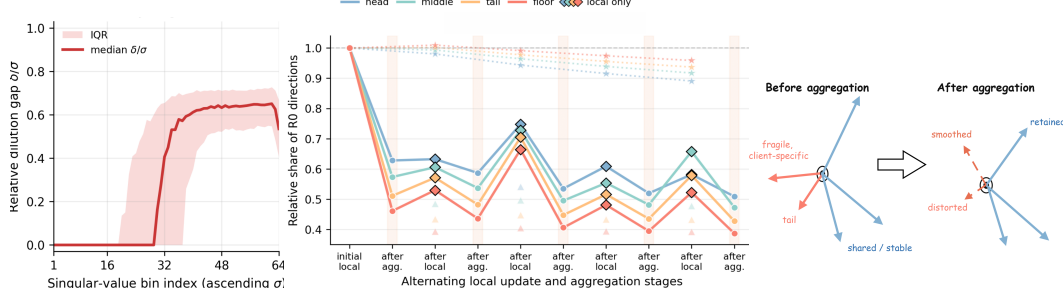


Figure 2: **Motivation of FedReCall.** Factor-wise LoRA aggregation repeatedly dilutes fragile client-specific directions: (a) the relative gap grows toward tail directions, (b) local training recovers them before aggregation suppresses them again, and (c) shared directions survive while private ones require client-side recall.

3.2 Motivation: directional dilution under factor-wise aggregation

Each direction ϕ_i in a client’s update carries a portion of its adaptation signal, weighted by its amplitude σ_i . Under factor-wise averaging (Eq. 3), the resulting global update does not preserve these client directions equally: large-amplitude (head) directions are largely shared across clients and survive aggregation; small-amplitude (tail/floor) directions, by contrast, encode client-specific signal; their lower amplitude leaves them more vulnerable to bilinear distortion—an asymmetry also documented in the model-merging literature [41].

To measure this vulnerability precisely, we define two per-direction quantities. The *global retention* of direction ϕ_i is

$$\alpha_i = \langle \Delta W_l^{t, \text{glob}}, \phi_i \rangle_F = s_l u_i^\top B_l^{t, \text{glob}} (A_l^{t, \text{glob}} v_i), \quad (5)$$

and the *dilution gap* is the local strength missing from this projection,

$$\delta_i = \max(0, \sigma_i - \alpha_i^+), \quad \alpha_i^+ = \max(\alpha_i, 0). \quad (6)$$

The relative gap $\delta_i/\sigma_i \in [0, 1]$ serves as a scale-invariant diagnostic; the absolute gap δ_i will be reused as the cache amplitude in Section 3.3. We make two observations: *which* directions are most affected (Obs. 1), and *what role* they play (Obs. 2).

Observation 1: Dilution is large and concentrated in the lower bands. Figure 2 provides a population-level view of the diagnostic: across all 80,640 direction instances, δ/σ is near zero at low absolute σ and rises smoothly toward the highest σ percentiles. This trend suggests that the measured gap is not confined to the smallest singular directions. Grouping by within-client band (Figure 2), the median δ_i/σ_i rises monotonically from head to floor; the lower a direction sits in its owner’s ranking, the larger the share of its strength removed by factor-wise averaging.

Observation 2: The diluted directions encode client-specific adaptation knowledge. Three independent lines of evidence show that the directions most affected by aggregation carry functionally meaningful, client-owned signal. (i) *Persistence under local training.* When aggregation is turned off (dashed lines in Figure 2), the band ordering *reverses*: tail and floor directions are preserved best, while head and middle decay fastest. Since SGD on a client’s own data retains whatever actively reduces the local loss, the persistence of tail/floor directions confirms that they contribute to the client’s task performance. (ii) *Owner specificity.* Triangles in Figure 2 show that a direction defined by client c is recovered better by c in subsequent rounds than by any other client, with the specificity gap largest in tail/floor. These directions therefore encode adaptation signal that is client-specific rather than population-shared. (iii) *Loss-blind validation.* Section 4.3 provides an out-of-sample test that further isolates this functional role: a single-batch loss probe with no access to direction labels accepts $\sim 60\%$ of cache injections built from tail/floor directions (loss decreases on a held-out batch) and rejects 100% of cache injections built from head/middle directions (loss increases). This direction-blind asymmetry directly shows that restoring a tail/floor direction reduces loss for its owner client, while restoring a head/middle direction does not.

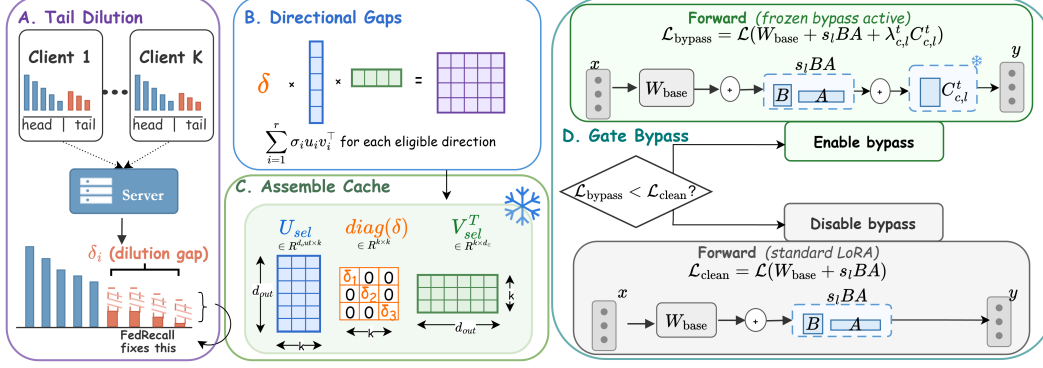


Figure 3: **Overview of FedReCall.** Factor-wise aggregation creates dilution gaps in weak client-specific directions. After aggregation, each client estimates the missing directional strength, assembles the reliable gaps into a private frozen cache, and reuses the cache as a low-rank bypass in later local training. A one-batch gate enables the bypass only when it lowers the client’s local loss; otherwise the client falls back to standard LoRA.

Together, these observations show that factor-wise aggregation systematically destroys the tail/floor directions encoding client-specific adaptation. The persistent zigzag in Figure 2 further shows that this knowledge loss is not a one-time event: the same directions are re-learned every round only for the next aggregation to erase them again, creating an avoidable recovery cost at every participation. Rather than redesigning the server aggregator—which would risk disrupting the cross-client consensus that preserves head directions—we keep these diluted directions as private memory on the clients that originated them.

3.3 FedReCall: client-private directional recall

Guided by the dilution analysis above, FedReCall protects diluted tail/floor directions as client-private memory rather than as shared parameters. It is a personalization plugin on top of standard factor-wise FedIT: clients still upload only clean LoRA factors, and the server aggregation rule is unchanged. After each aggregation a participating client builds a correction cache (**cache construction**); when next selected, it loads the cache as a frozen bypass during local adaptation (**bypass injection**), gated against stale caches by a one-batch loss probe (**quality gate**).

Cache construction. After client c completes local training in round t and receives the new post-aggregation global factors $(A_l^{t+1, \text{glob}}, B_l^{t+1, \text{glob}})$, it builds the correction cache in three steps. First, the client decomposes its *effective update* via SVD to obtain the singular triplets $\{(\sigma_i, u_i, v_i)\}_{i=0}^{r-1}$. The effective update captures both the trained LoRA factors and any active bypass from the current round,

$$\widetilde{\Delta W}_{c,l}^t = s_l B_{c,l}^t A_{c,l}^t + m_c^t \lambda_{c,l}^{\text{prev}} C_{c,l}^{\text{prev}}, \quad (7)$$

where $m_c^t \in \{0, 1\}$ indicates whether the bypass was enabled in round t and $C_{c,l}^{\text{prev}}$ is the cache from the client’s most recent prior participation; for a first-time client, $m_c^t = 0$ and the effective update reduces to the trained LoRA factors alone. Including the active bypass in the effective update provides a natural self-correction mechanism across rounds: if a cached direction has been absorbed by the global model, its new δ_i shrinks and the direction exits the cache, while directions that remain absent from the global model persist with large δ_i . Second, it computes the global retention α_i and dilution gap δ_i (Eqs. 5–6) for each direction, using the Frobenius projection defined in Section 3.2; the value δ_i estimates how much client signal is missing from the current global model along direction ϕ_i and becomes the amplitude stored in the cache. Third, motivated by the finding that tail/floor directions are both more strongly diluted and more client-specific (Section 3.2), it selects the eligible tail/floor set:

$$I_{c,l}^{t, \text{elig}} = \{i : \delta_i > 0, \sigma_i \geq \tau_l, i \in \text{tail} \cup \text{floor}\}, \quad (8)$$

where $\tau_l = \epsilon_{\text{rank}} \sigma_{l, \text{max}}$ with $\epsilon_{\text{rank}} = 10^{-4}$ excludes near-zero directions, and the index i is restricted to the tail and floor bands defined in Section 3.1. For LoRA rank $r = 32$, this restriction picks up

indices $i \geq 20$, yielding up to 12 candidate directions per layer. The private correction cache for layer l is

$$C_{c,l}^t = \sum_{i \in I_{c,l}^{t,\text{elig}}} \delta_i u_i v_i^\top. \quad (9)$$

Because the eligible set spans the tail and floor bands, $\text{rank}(C_{c,l})$ is bounded by the number of tail/floor indices. With $r = 32$, this yields at most 12 triplets (δ_i, u_i, v_i) per layer, requiring $12(d_{\text{out}} + d_{\text{in}} + 1)$ scalars—negligible relative to the LoRA factors themselves. The SVD is computed efficiently via QR-compressed low-rank algebra without materializing the full $d_{\text{out}} \times d_{\text{in}}$ matrix (Appendix B).

Bypass injection and calibration. When client c is next selected, it loads the cached correction and injects it as a frozen additive bypass in each layer’s forward pass:

$$y = W_{\text{base}}x + s_l B_l A_l x + \lambda_{c,l} C_{c,l} x. \quad (10)$$

During local training, only A_l and B_l are optimized; $C_{c,l}$ is frozen and excluded from the optimizer, and the gradient pathways to A_l and B_l are unchanged (Appendix C). The bypass exploits the low-rank structure of $C_{c,l}$:

$$C_{c,l} x = U_{\text{sel}} [\text{diag}(\delta) (V_{\text{sel}}^\top x)], \quad (11)$$

where U_{sel} and V_{sel} stack the selected left and right singular vectors. The bypass strength is calibrated relative to the clean LoRA update norm via a global hyperparameter η :

$$\lambda_{c,l} = \eta \cdot \frac{\|s_l B_l A_l\|_F}{\|C_{c,l}\|_F + \epsilon}. \quad (12)$$

Because the selected direction atoms are Frobenius-orthonormal, $\|C_{c,l}\|_F = \|\delta\|_2$, which can be computed without materializing the full correction matrix. This calibration ensures that the bypass-to-LoRA norm ratio $\|\lambda_{c,l} C_{c,l}\|_F / \|s_l B_l A_l\|_F \approx \eta$ holds consistently across layers and clients regardless of their absolute scales (derivation in Appendix D). Intuitively, η controls the relative influence of private memory versus the shared LoRA update during local adaptation: $\eta = 0$ recovers standard federated LoRA (no bypass), while larger η amplifies the correction. Because the norm calibration in Eq. 12 adapts $\lambda_{c,l}$ to each layer’s local scale, a single global η suffices without per-layer tuning; the sensitivity of η is studied in Section 4.3. The cache is never uploaded; the server receives only the clean trained LoRA factors $(A_{c,l}^t, B_{c,l}^t)$.

One-batch quality gate. A client’s cached correction may become stale if the global model has evolved substantially since the cache was built. Before enabling the bypass, FedReCall evaluates a single local mini-batch $\mathcal{B}_{\text{gate}}$ under both the clean global model and the bypassed model:

$$m_c^t = \mathbf{1}[\mathcal{L}(\theta^{\text{LoRA}}, \Lambda_c C_c; \mathcal{B}_{\text{gate}}) < \mathcal{L}(\theta^{\text{LoRA}}, \mathbf{0}; \mathcal{B}_{\text{gate}})], \quad (13)$$

where $\theta^{\text{LoRA}} = \{(A_l, B_l)\}_l$ denotes the shared LoRA factors, $\Lambda_c C_c = \{\lambda_{c,l} C_{c,l}\}_l$ denotes the per-layer frozen bypass, and $\mathbf{0}$ indicates the bypass is disabled. If the gate rejects the cache, the client trains with standard LoRA for that round. The gate’s overhead is a single forward pass per model variant.

The full per-round procedure is summarized in Algorithm 1; beyond standard federated LoRA, FedReCall adds only the client-side cache construction (Phase 3) and bypass injection (Phase 1), with the server aggregation unchanged and a single hyperparameter η .

4 Experiments

4.1 Experimental setup

Following Bai et al. [2], we construct a task-heterogeneous federated setting from the Super-Natural Instructions (SuperNI) dataset [33]. The full pool contains 511 NLP tasks from 76 task types; each

Algorithm 1 FedReCall: client c , round t

Require: $\eta > 0$

```

1: Input:  $\{A_l^t, B_l^t\}_l, \mathcal{D}_c$ ; optional  $C_c^{\text{prev}}$ 
2: // Phase 1: Bypass injection
3: if  $C_c^{\text{prev}}$  exists then
4:    $\lambda_{c,l} \leftarrow \eta \cdot \|s_l B_l^t A_l^t\|_F / (\|C_{c,l}^{\text{prev}}\|_F + \epsilon)$ 
5:    $m_c^t \leftarrow \mathbf{1}[\mathcal{L}(\theta, \lambda C; \mathcal{B}) < \mathcal{L}(\theta, \mathbf{0}; \mathcal{B})]$ 
6:   if  $m_c^t = 1$  then
7:      $y \leftarrow \lambda_{c,l} C_{c,l}^{\text{prev}} x$  (frozen)
8:   end if
9: else
10:   $m_c^t \leftarrow 0$  (bootstrap)
11: end if
12: // Phase 2: Local training
13:  $\{(A_{c,l}^t, B_{c,l}^t)\}_l \leftarrow \text{SGD}(\{A_l, B_l\}_l; \mathcal{D}_c)$ 
14: Upload  $\{(A_{c,l}^t, B_{c,l}^t)\}_l$ 
15: // Phase 3: Cache construction
16: Receive  $\{A_l^{t+1}, B_l^{t+1}\}_l$ ; compute  $\widehat{\Delta W}_{c,l}^t$ 
17:  $\alpha_i \leftarrow \langle \Delta W_l^{t+1, \text{glob}}, \phi_i \rangle_F$ ;  $\delta_i \leftarrow \max(0, \sigma_i - \alpha_i^+)$ 
18:  $I_{c,l}^t \leftarrow \{i : \delta_i > 0, i \in \text{tail} \cup \text{floor}\}$ 
19:  $C_{c,l}^t \leftarrow \sum_{i \in I_{c,l}^t} \delta_i u_i v_i^\top$ ; store locally

```

Table 2: Main performance on the task-heterogeneous federation.

Method	Personalized client model			Clean global model		
	Avg Pers. Loss ↓	Worst Loss ↓	Pers. ROUGE-L ↑	Avg Loss ↓	ROUGE-L ↑	Pers. Gap
FedIT	1.225	2.201	70.63	2.252	61.04	0.929
+ FedReCall	1.107	2.130	74.04	2.029	66.41	0.922
FLoRA	1.273	1.4598	70.85	2.397	67.50	1.124
+ FedReCall	1.146	1.4059	82.48	1.807	69.44	0.661
FFA-LoRA	1.345	1.960	56.72	1.985	71.28	0.640
+ FedReCall	1.089	1.554	84.47	1.244	80.58	0.155
FedSA-LoRA	1.291	2.160	69.70	2.045	62.33	0.754
+ FedReCall	1.236	2.168	75.27	1.414	77.16	0.178
<i>Dual-LoRA personalized baselines</i>						
FedDPA-T (1×)	1.809	2.035	57.51	2.111	57.93	0.302
FedDPA-T (2×)	1.177	2.001	73.25	2.390	51.64	1.213
FDLoRA (1×)	1.326	2.028	70.45	2.414	64.56	1.088

client holds a single task. We randomly draw 50 clients (seed 42) as the training federation; details of the client selection are provided in Appendix E. This federation is intentionally heterogeneous: each client corresponds to a distinct SuperNI task, the selected 50-client cohort spans 23 task types, and client data volumes are highly uneven. Thus the reported losses reflect a task-heterogeneous instruction-following setting rather than a balanced single-task benchmark. The base model is DataJuicer-LLaMA-1.3B [32], fine-tuned with LoRA [12] ($r=32$); all methods share the same rank for fair comparison.

We compare against two families of baselines. (i) *Aggregation-side methods*—**FedIT** [23], **FLoRA** [34], **FFA-LoRA** [30], and **FedSA-LoRA** [4]—differ in how they aggregate LoRA factors; FedReCall is layered on top of each as a plug-in (Table 2). (ii) *Dual-LoRA personalization methods*—**FedDPA-T** [38] and **FDLoRA** [26]—each maintain a global and a local adapter; they serve as comparison points for the personalization capability of FedReCall.

Personalized metrics (Avg Pers. Loss, Avg Pers. ROUGE-L) measure each client’s post-local-adaptation performance. *Global metrics* (Avg Loss, Avg ROUGE-L) assess the broadcast model before any local adaptation, tracking how the plug-in influences the shared model.

4.2 Main results: FedReCall as a plug-in

Table 2 reports all baselines on the 50-client SuperNI-derived federation, with FedReCall applied as an additive plug-in on top of each aggregator. We additionally report the personalization gap, defined as clean global loss minus average personalized loss; it contextualizes local gains by checking whether personalization is obtained on top of, rather than at the expense of, a useful global model.

FedReCall consistently improves personalized adaptation as a plug-in. Existing federated-LoRA aggregators face an inherent limitation: factor-wise averaging dilutes client-specific directions, and the severity depends on each aggregator’s structural constraints. Matrix-level protection offers only a partial solution because it makes a coarse decision about an entire factor. FFA-LoRA stabilizes aggregation by freezing one factor, but this restriction weakens client adaptation. FedSA-LoRA keeps the entire B factor local, which avoids uploading B -side private signal but also preserves all B -side content indiscriminately, including directions that may be stale, weak, or irrelevant for the current client. FedReCall is more selective: it keeps both LoRA factors fully trainable and shared, while recalling only directions with an observable dilution gap and a positive gate decision. As a plug-in that preserves the trainable-parameter set and adds only a bypass-strength hyperparameter η , FedReCall reduces personalized loss by 4.2%–19.0% across all four aggregators (Table 2). The larger gains on methods with more restrictive factor policies suggest that FedReCall is most useful when aggregation leaves more room for directional repair; Section 4.3 tests this interpretation through the η sensitivity and fragility diagnostics.

Explicit PFL-style compensation also comes with a different cost profile. FedDPA-T and FDLORA maintain separate global and local adapters, so personalization is obtained by allocating training capacity to an additional local module and, in FedDPA, by fusing the global and local adapters at inference. FedDPA-T ($1\times$) uses the same local training-step budget as the other methods; under this matched budget, splitting optimization across local and global adapters leaves both adapters undertrained in our setting. FedDPA-T ($2\times$) doubles the local optimization budget and narrows the gap, but still does not match FedReCall. The strongest FedReCall variant therefore outperforms both the matched-compute and doubled-compute dual-adapter baselines while preserving the original single-LoRA training protocol.

FedReCall simultaneously strengthens the global model, breaking the baseline trade-off.

FedReCall also improves global model quality across all four aggregators, despite its cache never being uploaded—the server-side aggregation is mechanically identical to the base method. The global gain therefore cannot come from the server directly receiving private cached directions. We interpret it as a change in the burden placed on the next round of uploaded clean LoRA factors: once recurring private corrections are recalled locally, the trainable factors can carry more shared adaptation signal while the non-shared correction remains local. The effect is most pronounced on FFA-LoRA, where global ROUGE-L rises by +9.3 points and global loss drops by 37%. This dual improvement—personalized *and* global—makes FedReCall the only method in Table 2 that reaches the upper-right corner of strong performance on both axes. FedReCall introduces only modest client-side computation for cache construction and gate evaluation; detailed wall-clock measurements are reported in Appendix I.

FedReCall preserves clean global generalization.

Table 3 evaluates the final broadcast model on held-out task types without local adaptation or private recall. This setting isolates the clean global model from the client-side bypass. The results show that the improved global metrics in Table 2 are not an artifact of evaluating with private memory enabled: the final broadcast model remains competitive on held-out task types while the cache stays strictly local.

4.3 Ablation and analysis

We next test whether FedReCall’s design choices are necessary for repairing directional dilution rather than merely adding extra local capacity. The ablations follow the mechanism in Section 3.2: a useful correction should restore diluted directions locally, reject stale private signal, and operate on the effective update ΔW where dilution is defined.

Component ablation.

The frozen bypass and the quality gate play different roles. The bypass supplies the missing directions during local adaptation, which accounts for most of the personalized improvement. The gate turns this memory into a validated correction: a cached direction is used only when it improves the current client’s mini-batch loss. This matters because early caches can be stale or over-amplified; without the gate, recall remains helpful for personalization but can degrade the shared model.

Table 4: Component ablation of the private recall mechanism.

Bypass	Gate	Avg. Pers. Loss ↓	Global Loss ↓	Δ vs. FedIT
–	–	1.3235	2.2528	–
–	–	1.073	2.0297	–16.3%

[†]The bypass-only run uses an alternate seed because the default run diverged before the final round.

Training dynamics.

Figure 4 traces the per-round personalized and global loss across all compared methods. Baselines without FedReCall show round-to-round oscillations consistent with the recovery–disruption cycle in Section 3.2; FedReCall dampens these oscillations and converges to lower loss.

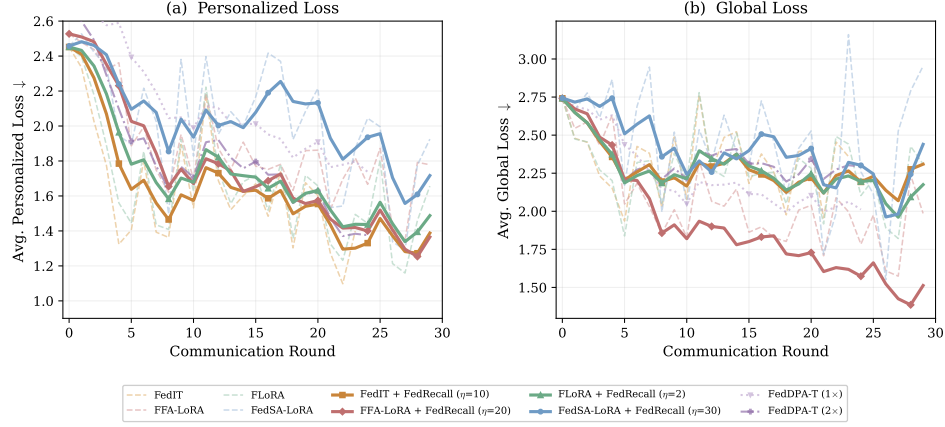


Figure 4: **Training dynamics with and without FedReCall.** Dashed curves are base aggregators; solid curves add FedReCall, with EMA smoothing used only for visualization.

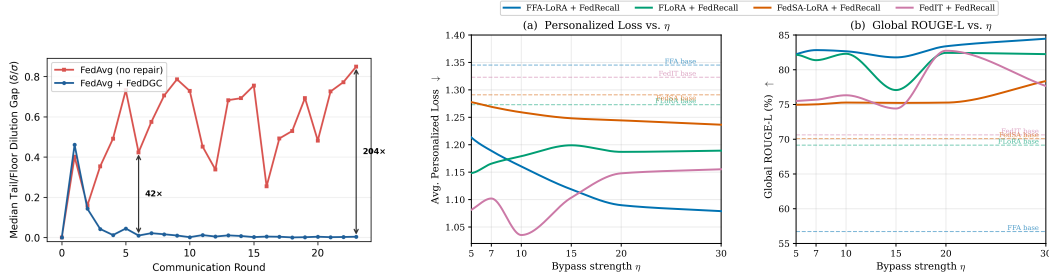


Figure 5: **Fragility repair diagnostic.** The relative dilution gap is tracked after recall is enabled.

Figure 6: **Bypass strength sensitivity.** The useful η range depends on the base aggregator.

Bypass strength sensitivity. Figure 6 studies the bypass strength η , which controls how strongly the cached directions are recalled relative to the current LoRA update. This experiment is not intended to show a linear relationship between η and performance. Rather, it tests whether the norm-calibrated bypass in Eq. 12 requires a finely tuned scaling factor. Across base aggregators, FedReCall remains effective over a non-trivial range of η values, indicating that the recall mechanism is not tied to a narrow optimum. The preferred range still depends on the underlying aggregation rule: methods that leave stronger directional dilution benefit from stronger recall, whereas methods that already preserve more private signal peak at a more moderate compensation strength. Thus η is best viewed as a coarse compensation knob: it simplifies hyperparameter search by using one global scaling factor across layers and clients, but overly large values can still over-trust stale cached directions.

Fragility repair. The fragility diagnostic closes the loop with the motivation. It applies the same relative dilution measure used to identify vulnerable directions, but after FedReCall is enabled. The diagnostic moves toward zero across rounds, indicating that the bypass repairs the specific directional gap created by factor-wise aggregation. This distinguishes FedReCall from a generic local-capacity method: it improves the mechanism-level failure signal, not only the downstream loss.

Gate behaviour. The quality gate is deliberately direction-blind: it compares the local loss with and without the bypass and does not know whether a cache was built from a particular direction-strength range. Its behaviour therefore tests whether the low-strength prior is functionally meaningful rather than hard-coded into the gate. Table 5 shows that low-strength candidates reduce the gate loss and are frequently accepted, while high/mid-strength candidates sharply increase the loss and are rejected. At runtime, FedReCall therefore uses the low-strength prior only to propose a cache; the loss gate functionally validates whether that cache should be used.

Table 5: Gate behaviour under direction-strength candidate pools.

Candidate range	Gate pass rate	Clean loss	Bypassed loss
Low-strength range ($q_i \geq 0.6$)	0.78 ± 0.19	2.00 ± 0.04	1.95 ± 0.04
Low-strength range, uniform amplitude	1.00 ± 0.00	2.07 ± 0.10	1.98 ± 0.07
High/mid-strength range ($q_i < 0.6$)	0.00 ± 0.00	1.96 ± 0.11	11.51 ± 0.33

SVD operand. The final ablation asks where the SVD should be applied. A factor-only cache is tempting because some FedLoRA variants explicitly share or freeze a single factor. However, the dilution gap is defined in the model’s effective update, not in either factor alone. When the same cache pipeline is applied to B alone, the benefit disappears; when it is applied to $\Delta W = BA$, the repair reappears. Table 6 in Appendix A reports this operand ablation and supports the rule that FedReCall should decompose the layerwise effective update, regardless of how a base aggregator partitions the LoRA factors.

5 Conclusion

Our analysis and experiments point to a common mechanism behind personalization loss in federated LoRA: under task heterogeneity, factor-wise aggregation repeatedly weakens locally useful directions that are weakly shared across clients. FedReCall turns this observation into a client-side repair mechanism. It estimates observable dilution gaps after aggregation, stores reliable missing directions in a private frozen cache, and recalls them through a gated low-rank bypass during later local adaptation. Because the cache is neither uploaded nor optimized, FedReCall leaves the server aggregation rule, communication payload, and trainable-parameter set unchanged. Across representative FedLoRA aggregators, the results show that selectively restoring diluted client-specific directions can improve personalized adaptation while maintaining global performance in our setting.

Limitations. We have not yet evaluated FedReCall on a broader range of datasets and downstream tasks. This leaves open how well the same directional-recall principle generalizes beyond the task-heterogeneous LoRA instruction-tuning setting studied here. Finally, although the cache is never uploaded, it remains client-specific state. Studying its interaction with formal privacy guarantees, secure local storage, other adapter families, and larger model scales is an important direction for future work.

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A Operand ablation

Table 6: Operand ablation on FFA-LoRA. Applying FedReCall to the effective update $\Delta W = BA$ yields clear average-loss and ROUGE-L gains, whereas a naive B -only variant provides no benefit.

Method	Weighted Loss ↓	ROUGE-L ↑	Worst Loss ↓
FFA-LoRA	1.3931	0.6838	2.6645
+ FedReCall (B -only, naive)	1.4050	0.6755	2.6712
+ FedReCall ($\Delta W = BA$, principled)	1.0834	0.7723	2.7064

B QR-compressed SVD for cache construction

The cache construction requires the SVD of the effective update $\widetilde{\Delta W}_{c,l}^t \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ (Eq. 7), which can be prohibitively expensive if materialized as a dense matrix. We exploit the low-rank factored form to avoid this.

When no bypass is active ($m_c^t = 0$), the effective update is $s_l B_{c,l}^t A_{c,l}^t$, which already has rank at most r . When the bypass is active, we have

$$\widetilde{\Delta W}_{c,l}^t = \underbrace{[s_l B_{c,l}^t \mid \lambda_{c,l}^{\text{prev}} U_{\text{sel}} \text{diag}(\delta^{\text{prev}})]}_{L \in \mathbb{R}^{d_{\text{out}} \times (r+k)}} \underbrace{\begin{bmatrix} A_{c,l}^t \\ V_{\text{sel}}^{\text{prev}\top} \end{bmatrix}}_{R \in \mathbb{R}^{(r+k) \times d_{\text{in}}}},$$

where $k = |I_{c,l}^{\text{prev}}|$ is the number of cached tail/floor directions (at most $|\text{tail}| + |\text{floor}|$). The product LR has rank at most $r + k$, and its SVD can be obtained without forming the $d_{\text{out}} \times d_{\text{in}}$ matrix:

1. Compute a thin QR factorization $L = Q_L R_L$, where $Q_L \in \mathbb{R}^{d_{\text{out}} \times (r+k)}$ has orthonormal columns and $R_L \in \mathbb{R}^{(r+k) \times (r+k)}$.
2. Form the small matrix $M = R_L R \in \mathbb{R}^{(r+k) \times d_{\text{in}}}$.
3. Compute the SVD $M = U_M \Sigma V^\top$.
4. The left singular vectors of $\widetilde{\Delta W}_{c,l}^t$ are $Q_L U_M$; the singular values and right singular vectors are Σ and V .

All operations involve matrices of size at most $(r + k) \times \max(d_{\text{out}}, d_{\text{in}})$, giving a total cost of $O((r + k)^2(d_{\text{out}} + d_{\text{in}}))$ per layer, compared to $O(d_{\text{out}} d_{\text{in}} \min(d_{\text{out}}, d_{\text{in}}))$ for a dense SVD.

C Gradient flow through the frozen bypass

A natural concern is whether the frozen bypass distorts the gradient signal received by the trainable LoRA factors. We show that it does not block or redirect any gradient pathway.

The bypassed forward pass for layer l is (Eq. 10)

$$y = W_{\text{base}}x + s_l B_l A_l x + \lambda_{c,l} C_{c,l} x.$$

Because $C_{c,l}$ is a frozen constant, the partial derivatives of y with respect to the trainable factors are

$$\frac{\partial y}{\partial B_l} = s_l (A_l x) \otimes I, \quad \frac{\partial y}{\partial A_l} = s_l B_l^\top \otimes x^\top,$$

which are identical to the standard LoRA case without bypass. The bypass term $\lambda_{c,l} C_{c,l} x$ affects the loss $\mathcal{L}$ by changing the output y , and therefore changes the upstream gradient $\partial \mathcal{L} / \partial y$. However, the Jacobians $\partial y / \partial B_l$ and $\partial y / \partial A_l$ are bypass-independent.

In effect, the frozen bypass shifts the output baseline from which LoRA operates: during local adaptation, the LoRA factors need not spend capacity re-learning the directions already restored by the cache, and can focus on further task-specific adaptation. No stop-gradient is applied to the shared input x , so gradient information flows freely to all layers below.

D Bypass strength calibration derivation

We show that the per-layer scaling $\lambda_{c,l}$ in Eq. 12 achieves the target bypass-to-LoRA norm ratio η up to negligible ϵ -stabilizer terms.

Setup. The cache admits the factorization $C_{c,l} = U_{\text{sel}} \text{diag}(\boldsymbol{\delta}) V_{\text{sel}}^\top$, where the columns of U_{sel} and V_{sel} are subsets of orthonormal singular vectors. Because distinct direction atoms $\phi_i = u_i v_i^\top$ are Frobenius-orthogonal,

$$\|C_{c,l}\|_F^2 = \sum_{i \in I_{c,l}^{\text{elig}}} \delta_i^2 = \|\boldsymbol{\delta}\|_2^2.$$

Derivation. Substituting Eq. 12 into the bypass norm:

$$\|\lambda_{c,l} C_{c,l}\|_F = \lambda_{c,l} \|C_{c,l}\|_F = \eta \cdot \frac{\|s_l B_l A_l\|_F}{\|C_{c,l}\|_F + \epsilon} \cdot \|C_{c,l}\|_F.$$

Dividing by $\|s_l B_l A_l\|_F$:

$$\frac{\|\lambda_{c,l} C_{c,l}\|_F}{\|s_l B_l A_l\|_F} = \eta \cdot \frac{\|C_{c,l}\|_F}{\|C_{c,l}\|_F + \epsilon}.$$

When $\|C_{c,l}\|_F \gg \epsilon$ (which holds for any non-trivial cache), the ratio approaches η . The ϵ term (set to 10^{-8} in practice) serves only as a numerical guard against division by zero and does not affect the calibration for any layer with a non-empty cache.

E Data preparation and client selection

Client pool. Following Bai et al. [2], we use Super-Natural Instructions [33] to construct a task-heterogeneous federation of 511 clients covering 76 NLP task types. Each client holds a single task, and within-task training data is subsampled to 10% following the original FlexLoRA protocol.

50-client subset. From the 511-client pool we draw 50 clients uniformly at random (`random.sample(range(511), 50, seed=42)`) and re-index them as clients $0, \dots, 49$. A deterministic mapping from new to original client IDs is stored for reproducibility. The selected 50 clients span 23 NLP task types; per-client training-set sizes range from 16 to 520 samples (median 127, mean 232).

Per-round participation. In each communication round, 5 clients (10%) are sampled without replacement. The client-selection seed is fixed to 42 and shared across all training seeds {42, 43, 44}, so that every run uses an identical participation schedule. This design isolates the reported variance to training randomness alone, excluding participation-schedule noise.

F Hyperparameters

G Bypass strength selection

Table 7 reports the full η sweep for all four base methods (seed 42, $q=0.6$, 30 rounds). For FedIT+FedReCall we select $\eta=10$ (best personalized loss 1.035). For FFA-LoRA+FedReCall we select $\eta=20$ (loss 1.090, -19% vs. baseline). For FedSA-LoRA+FedReCall we select $\eta=30$ (loss 1.237, -3.1% vs. baseline). For FLoRA+FedReCall we select $\eta=2$ (loss 1.147, -1.1% vs. baseline).

Table 7: Full bypass-strength sweep results.

η	2	3.5	5	7	10	15	20	30
FFA-LoRA	1.282	1.248	1.213	1.189	1.160	1.119	1.090	1.081
FLoRA	1.147	1.176	1.168	1.186	1.179	1.199	1.167	1.1892
FedSA-LoRA	1.285	1.282	1.278	1.269	1.259	1.248	1.244	1.237
FedIT	1.2823	1.2484	1.081	1.102	1.035	1.104	1.148	1.155

H Metric definitions

I Computational infrastructure

All experiments are conducted on a single NVIDIA GeForce RTX 5090 GPU. The codebase is built on PyTorch 2.7.0 (CUDA 12.8).

Per-round wall-clock time. Table 8 reports the average per-round wall-clock time measured over the stable training phase (excluding initial warm-up rounds).

Table 8: Per-round wall-clock time (minutes). *FedReCall overhead* is the additional time introduced by the plug-in over the corresponding base method.

Method	Per-Round (min)	vs. FedIT	FedReCall Overhead
FedIT	3.4	–	–
+ FedReCall	4.7	+38%	+1.3 min
FFA-LoRA	4.1	+21%	–
+ FedReCall	4.9	+44%	+0.8 min
FLoRA	5.5	+62%	–
+ FedReCall	6.5	+91%	+1.0 min
FedSA-LoRA	4.4	+29%	–
+ FedReCall	4.7	+38%	+0.3 min
<i>Dual-LoRA personalized baselines</i>			
FedDPA-T ($1\times$)	5.1	+50%	–
FedDPA-T ($2\times$)	6.3	+85%	–
FLoRA	4.8	+41%	–

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